

Final report: decoder root causes and inverse architecture

Executive conclusion

The poor auto-encoding result was not one undifferentiated “Perceiver problem.” The experiments isolate seven mechanisms, each with a direct intervention:

Cause	Decisive evidence	Architectural consequence
One-latent query blindness	Spatial output difference and query-gradient norm are both exactly zero	Do not use a one-latent Perceiver IO decoder for multiple output queries
Narrow attention value path	Widening the actual value path from 64 to 128 lowers held-out one-cell MSE from 0.239298 to 0.021848	Latent/query width alone is not the relevant capacity control
Unanchored spatial routing	Fixed fields can be memorized at $\sim 1e-3$ train MSE while held-out MSE is 1.660140 ; physical anchoring removes the scale sensitivity	Give output transport an explicit physical route
Encoder spatial bandwidth loss	Fixed coarse patches discard half-degree subcell structure; a learned native-grid encoder restores it	Keep learned state features on the native input grid
Wet-mask ordering	Projecting before per-channel masked resampling removes 78% of half-to-one excess error on identical weights	Decode physical channels before spatial transport
Boundary contamination	A state-only inverse lowers one-degree reconstruction by 78.5% and improves all four one/half-degree routes	Encode persistent state once; encode one aligned boundary state per processor call
Forecast head co-adaptation	Soft inverse loss leaves 18.7% encoder and 14.6% decoder drift; freezing holds reconstruction exactly	Freeze the learned inverse and train a zero-initialized latent residual transition

The selected architecture is therefore:

```
z_0      = E_state(x_t)                # learned, native-grid state
q_m      = E_boundary(b_m)             # one aligned forcing state
z_m      = z_(m-1) + alpha * P(z_(m-1), q_m, geometry)
native_y = pointwise_decode(z_m)       # raw values; no decoder LayerNorm
x_hat_s,m = channel_masked_resample(native_y, source_grid, output_grid)
```

`alpha` is initialized to zero. Latitude, longitude, and scale do not alter the reconstructive state tensor; they enter each processor call through a sidecar and the deterministic decoder route. The encoder is not initialized or constrained to be an identity. The learned inverse is the composition $D(E_state(x))$, and the same latent can be processed zero through `N` times before decoding at a requested resolution. A decoder attention residual is not selected: the learned-encoder screen found no accuracy or spectral defect that justified its 4–10x cost.

The full one-degree run validates the latent forecast architecture at {1,2,4}-step MSE {0.0205191, 0.0343797, 0.0469026} while keeping the frozen inverse exactly at 0.000647529. The full one/half-degree run then completes all 6,392 updates in 10h30m. Its validation-selected checkpoint is also terminal, with aggregate lead MSE {0.0398245, 0.0540801, 0.0659528}, respectively {62.9%, 69.5%, 73.9%} below lead-matched persistence. Same-grid reconstruction remains exactly 0.00111017 at one degree and 0.00124296 at half degree. Zeroing the per-step

boundary raises the three aggregate leads by {23.4%, 61.1%, 124.7%}; reversing boundary order raises them by {9.8%, 19.2%, 93.7%}. This is strong evidence that the separately encoded boundary path is used physically rather than ignored.

The endpoint spatial audit distinguishes decoder success from processor and information limits. Temperature, salinity, and SSH high-wavenumber ratios are 0.938--1.025 on every route. Fine-to-coarse velocity ratios are 0.884/0.872, and defined seam ratios span 0.811--1.002. Half-degree same-grid forecast velocity remains underpowered at 0.790/0.751, while one-to-half velocity is only 0.288/0.149; deterministic prolongation cannot invent modes absent from the coarse source. Because the inverse is frozen and unchanged, these forecast-only spectral limitations belong to the processor/exposure and source information, not to renewed decoder query blindness.

The evidence supports the selected encoder/decoder inverse through one and half degree, but it does not yet support a larger quarter-degree training run. One-to-one lead-one MSE in the multi-resolution model is 0.0264174, 28.7% above the single-resolution result and just outside the provisional 25% degradation gate. The fourfold quarter-to-one zero-shot audit also aliases under bilinear point restriction. Retain the selected inverse and latent-AR contract, then review whether to increase per-route processor exposure/capacity and validate the checked-in scale-aware conservative restriction before spending on quarter-degree training. That restriction is unit-tested but empirically unvalidated and is not part of the selected architecture in this report.

Objective

Isolate why the SamudraMulti Perceiver IO decoder performs poorly on identity reconstruction, then identify architectural changes that preserve flexible output resolutions. This report combines the completed real-ocean experiments with new decoder-only controls; follow-up experiments are explicitly separated from the evidence supporting the conclusions.

Glossary and implementation map

This section defines the experimental shorthand used below. “Selected” means part of the recommendation validated by the one/half-degree run. “Retained control” means the implementation remains available for comparison but is not part of that recommendation. In the equations, E_s is the state encoder, E_b the boundary encoder, P the shared processor, D the decoder, z_m the latent after m physical steps, and R a deterministic spatial resampler.

Selected architecture and training contract

- **Frozen learned inverse (selected).** “Inverse” means the trained composition $D(E_s(x)) \approx x$; it does **not** mean that either module is an algebraic inverse, that their weights are tied, or that the encoder is an identity. “Frozen” means that after loading the reconstruction checkpoint, parameters whose names begin with `encoder.` or `decoder.` have `requires_grad=False`. Forecast training optimizes the processor, boundary encoder, geometry sidecar, and residual scale without changing E_s or D . The controls are `finetune: true`, `resume_ckpt_path`, and `frozen_model_prefixes: ["encoder.", "decoder."]` in the [selected forecast config](#); the exact prefix matching and freezing is in `freeze_model_parameters`.

- **Depth zero / zero-step reconstruction / S0 (selected invariant).** Encode and decode without calling the processor: $\hat{x}^{(0)} = D(E_s(x))$. This is the operation measured when the report quotes reconstruction or “inverse” MSE. The implementation is `SamudraMulti.reconstruct_once`. `bypass_processor: true` and `processor_iterations: 0` are reconstruction-run diagnostics; they are not required to call `reconstruct_once`.
- **State-only encoder (selected).** E_s receives prognostic state channels only. Boundary forcing is deliberately discarded by `SamudraMulti.encode`, so the latent that the decoder must reconstruct is not a function of transient forcing. Prognostic and boundary variable sets are selected independently by `data.dataset.prognostic_vars_key` and `boundary_vars_key` in the `training config`.
- **Native-grid learned projection encoder (selected).** With `encoder.native_projection: true`, every source cell is mapped independently, $z_{cij} = \sum_k W_{ck}^E x_{kij} + b_c^E$, and the source $H \times W$ grid is retained. There is no pooling or resampling in this encoder. The flag is in the `selected model config`; it builds `DirectPatchEncoder` with its one-cell-patch restriction disabled by `EncoderConfig.build`.
- **Geometry sidecar (selected).** `encoder.geometry_mode: "sidecar"` means no position or scale embedding is added to $E_s(x)$. Before each processor invocation, the model instead adds $W_g[\mathbf{r}(\phi, \lambda), \log A(\phi, \lambda)]$, where $\mathbf{r}$ is the three-dimensional unit-sphere coordinate and A is spherical cell area. The 1×1 map W_g is initialized to zero, so enabling the sidecar is initially an exact no-op. See the `selected flag`, `ProcessorGeometryConditioner`, and its construction in `SamudraMultiConfig.build`. The other retained modes are `"none"` (geometry enters neither encoder nor processor) and `"additive"` (learned position/scale terms are added directly to the encoder representation).
- **Separate, time-aligned boundary encoder (selected).** One boundary state is used per physical transition: $q_m = R_{\ell_b \rightarrow \ell_z}(W_b b_m)$, where W_b is a shared 1×1 projection and R is physical-coordinate bilinear resampling when the boundary and latent grids differ. It is added only to the corresponding processor input. See `BoundaryEncoder` and its use in `SamudraMulti.process`. There is no separate enable flag; it is constructed for non-bypassed `SamudraMulti` models from `boundary_vars_key`.
- **Latent autoregression / encode once (selected).** The physical state is encoded once, advanced entirely in latent space, and decoded only at requested leads. Depth N consumes $b_0, \dots, b_{N-1}$ and is trained against the state at $t + N dt$, not against $t + dt$ after N arbitrary network passes. `train_processor_depths: [1, 2, 4]` cycles training leads and `validation_processor_depths: [1, 2, 4]` audits all three; see the `selected training controls` and `SamudraMulti.latent_forecast`.
- **Latent ReZero transition / zero-initialized latent residual scale (selected).** Here “ReZero” refers to a residual **processor**, not to decoder attention:

$$z_m = z_{m-1} + \alpha \odot P(z_{m-1} + E_b(b_m) + W_g g), \quad \alpha_c = 0 \text{ at initialization.}$$

`processor_residual: true` enables a learned per-channel α , so a newly attached processor initially predicts latent persistence and can be invoked zero through N times. See the [selected flag](#) and [SamudraMulti.process](#).

- **Projection-before-channel-masked coordinate resampling (selected decoder).**

First decode latent channels on their native grid, $u_i = W^D z_i + b^D$. Then, independently for each prognostic channel c , render output coordinate q as

$$\hat{x}_c(q) = \frac{\sum_i w_i(q) M_{ci} u_{ci}}{\max(\epsilon, \sum_i w_i(q) M_{ci})}.$$

$w_i(q)$ are physical-coordinate bilinear weights, longitude is periodic, latitude is edge-clamped, and M_{ci} is that channel’s source wet mask. The three flags are

`decoder.resample_projection: true`, `coordinate_resampling: true`, and `project_before_resample: true` in the [selected model config](#). The implementation is [ResampleProjectionDecoder](#) plus [coordinate_bilinear_resample](#). This selected path contains no LayerNorm: reconstructive values pass only through the learned 1×1 projection and deterministic resampler.

- **Flexible output resolution (selected property).** Decoder weights do not depend on output height or width. [SamudraMulti.decode](#) passes the requested latitude/longitude arrays to the resampler, so the same E_s, P, D can render any supplied regular physical grid. Same-grid decoding skips spatial transport and reduces exactly to the learned 1×1 channel map. “Flexible” does not imply that coarse input contains enough information to produce truthful fine-grid modes.

- **Zero-depth auxiliary loss (available, disabled in the selected forecast).**

`zero_depth_reconstruction_weight:  $\lambda_0$` adds $\lambda_0 \text{MSE}(D(E_s(x_t)), x_t)$ to forecast training using the source grid. Its implementation is [training_auxiliary_loss](#). It is `0.0` in the [selected model](#) because freezing E_s, D preserves the inverse more strictly than a soft penalty.

- **Common one-degree normalization statistics (selected data contract).**

Both one- and half-degree examples use the one-degree channel means and standard deviations, $x' = (x - \mu_{1^\circ}) / \sigma_{1^\circ}$. This prevents a change of resolution from also changing the numerical coordinate system in which the learned inverse operates. The two source entries are explicit in the [selected training config](#).

- **Physical-state residual prediction (disabled).** `pred_residuals: false` means the

decoder predicts $\hat{x}_{t+N}$, not $x_t + \widehat{\Delta x}$. This is independent of `processor_residual: true`, which is the selected residual operation in latent space. Both choices are visible in the [selected model config](#).

Retained alternatives and diagnostic controls

- **Perceiver IO decoder / widened Perceiver IO (retained controls).** Each output **spatial** query is made from latitude/longitude and emits the complete output-channel vector; the query does not embed a target channel. The final cross-attention value width is `decoder.perceiver.cross_heads * decoder.perceiver.cross_dim_head`, while

`decoder.queries_dim` is the query embedding width. The production class is `PerceiverDecoder`, the exposed capacity controls are in `PerceiverConfig`, and the matched widened example is `model_1cell_decoder_wide.yaml`.

- **Direct query-to-token attention (retained synthetic control).** This removes the Perceiver’s internal learned latent bank. Each output-position query cross-attends directly to source tokens, retains a query residual, passes through a feed-forward block, and is projected to all output channels. It is selected by `--architecture direct-cross-attention` in `probe_perceiver_decoder.py` and implemented by `DirectCrossAttentionIO`.
- **Position-query anchored attention (called “position-only anchored attention” below; retained control).** “Position-only” means the **query input** contains only the output coordinate; it does not mean routing ignores source content. The tested local form uses

$$\text{softmax}_i \left(q(\mathbf{r}_q)^\top k(z_i) / \sqrt{d} - \beta d_{\text{phys}}(q, i)^2 \right) v(z_i)$$

over a bounded source neighborhood. Thus it combines a physical-distance anchor with learned content keys and values. The runnable probe route is `--architecture anchored-cross-attention`, built by `CoordinateAttentionDecoder` from `LocalCoordinateAttentionCorrection`; its main controls are `--context-patches`, `--position-bias-strength`, `--cross-heads`, and `--cross-dim-head`.

- **Resampling + zero-initialized attention residual / hybrid decoder (retained fallback).** This computes $D(z) = D_{\text{resample}}(z) + C_\theta(z, \text{coordinates})$. The final projection of C_θ is zero-initialized, so the module starts exactly as the deterministic resampling base. `residual_normalize_values: false` leaves attention values on a raw-amplitude path; LayerNorm still forms keys and feed-forward features inside the correction, but reconstructive base values do not pass through it. See `model_iterable_inverse_hybrid.yaml`, `ResampleAttentionResidualDecoder`, and `LocalCoordinateAttentionCorrection`. It remains a fallback because the learned-encoder screen did not justify its measured cost.
- **Canonical-resampling encoder (retained control).** With `encoder.canonical_resampling: true`, the model first applies the learned pointwise channel map and then bilinearly moves every input onto the finest configured grid. Unlike native projection, this gives P one fixed latent grid but may interpolate the representation before processing. See the [example config](#) and `CanonicalResampleEncoder`.
- **Direct patch encoder/decoder (retained controls).** These are learned 1×1 channel maps with no attention. `encoder.direct_projection: true` and `decoder.direct_projection: true` require processor and physical grids to match and require the configured physical patch to be one cell. They are implemented by `DirectPatchEncoder` and `DirectPatchDecoder`. `native_projection` deliberately reuses the encoder class while relaxing only its one-cell-patch assertion.
- **Packed spatial-query encoder (retained, structurally incomplete control).** `encoder.spatial_query_shape: [h, w]` asks one Perceiver IO call to emit an ordered $h \times w$ set of vectors, then packs those vectors into the channel axis at one coarse patch center.

`encoder.spatial_query_channels` sets the width of each vector. The current pointwise decoder has no intra-patch coordinate with which to unpack those channel groups spatially, and a one-internal-latent Perceiver makes its output queries query-blind. See the fields in `EncoderConfig` and `SpatialQueryPerceiver`. It is not selected; a future version would need to unpack onto a finer physical grid or use multiple/directly anchored encoder tokens.

- **Perceiver encoder LayerNorm controls (retained diagnostics).**

`encoder.perceiver.normalize_input_context: false` removes LayerNorm before encoder input cross-attention, and `encoder.perceiver.normalize_encoder_output: false` removes the LayerNorm after latent pooling and before the output projection. These are information-path diagnostics supported only by the naive attention implementation; see `PerceiverConfig`. They do not affect the selected native projection encoder, which contains no LayerNorm.

- **Scale-aware conservative restriction (implemented prototype, not selected).**

When `decoder.conservative_restriction_min_ratio: r` is set and both axes are reduced by at least r , bilinear point sampling is replaced by mask-renormalized cell-overlap averaging. Latitude overlap is integrated in $\sin \phi$, and longitude overlap is periodic. The checked-in prototype uses $r = 3$ in

`model_iterable_inverse_native_masked_conservative_projection.yaml`; the math is implemented by `coordinate_conservative_resample`.

- **Boundary ablations (retained validation controls).** `validation_boundary_ablations` can replace each aligned forcing sequence with `zero`, `batch_shuffle`, or `time_reverse`. A forecast degradation demonstrates that the processor uses the correct boundary values or ordering rather than merely tolerating the boundary path. See the field definition in `TrainConfig` and the selected `zero/time-reverse audit`.

- **Validation-only and identity-eval-only runs (retained audit modes).**

`validation_only: true` loads an explicit forecast checkpoint, performs one validation pass, and writes no optimizer or checkpoint state; see `TrainConfig.validation_only`.

`identity_eval_only: true` is the analogous no-update mode in the identity harness; see `IdentityConfig`. These modes audit a checkpoint without silently continuing its training.

- **Removed fine-scale-query feature (not present).** `use_fine_scale_queries` was deleted before these experiments. No selected or retained config depends on it.

Measurement vocabulary

- **Route $a \rightarrow b$.** Input resolution a , requested output/target resolution b . For example, `1 -> 1/2` encodes one-degree state and renders a half-degree target.
- **Persistence.** The lead-matched no-dynamics baseline: resample x_t to the target grid and compare it with $x_{t+N \, dt}$. “Persistence reduction” is $1 - \text{MSE}_{\text{model}} / \text{MSE}_{\text{persistence}}$.
- **High-wavenumber power ratio.** Generated zonal Fourier power divided by target power, summed over the upper half of nonzero resolved wavenumbers. One is amplitude-matched, below one is spectrally underpowered, and above one is overpowered. See `high_wavenumber_power_ratio`.
- **Patch-seam jump ratio.** Mean absolute error jump across nominal patch boundaries divided by the corresponding within-patch jump. One means the boundary is no worse

than an ordinary neighboring-pixel edge; above one indicates a boundary-aligned artifact. See `patch_seam_jump_ratio`.

- **Query blindness.** Outputs are invariant to changing the output query. In the one-internal-latent Perceiver case, the final attention softmax has length one, so every spatial query receives the same attended value and the query gradient is zero.
- **Attention value-path width.** The number of scalar features transported by cross-attention, `cross_heads * cross_dim_head`; it is distinct from the encoder/processor channel width, learned-latent width, and query embedding width.
- **Oracle copy and deterministic floor.** An oracle-copy probe places target channels directly in decoder input, so it isolates rendering without testing a learned encoder. A deterministic floor is the error of the specified geometric resampler on the same source and target fields; it is a reference for information and interpolation error, not a universal lower bound for a learned dynamical model.

Prior evidence

The completed 32-sample pure-autoencoder factorial documented in the [single-step research results](#) localizes nearly all of the surprising error to the decoder. With the processor bypassed, direct/direct reaches MSE `0.012083`, Perceiver/direct reaches `0.025431`, direct/Perceiver reaches `0.279303`, and Perceiver/Perceiver reaches `0.285353`. Replacing the Perceiver decoder with bilinear resampling plus a pointwise projection subsequently reduces the paired one-degree forecast proxy from the original SamudraMulti result of `0.381735` to `0.051655`.

Those experiments demonstrate localization and provide a successful fallback, but they do not distinguish mathematical degeneracy, information capacity, spatial routing, optimization, and memorization. The controls below remove the encoder, processor, data loader, ocean mask, boundary fields, and bfloat16 arithmetic. They feed 128-channel processor-like features directly to the production `PerceiverDecoder`; the first 77 feature channels are independent standard-normal targets and the remaining channels are zero.

The reproducible harness is `scripts/probe_perceiver_decoder.py`.

Root cause 1: a one-latent, multi-query decoder is query-blind

`perceiver-pytorch==0.8.8` performs the final decoder cross-attention without a query residual. With one internal latent, the attention softmax has length one and is identically one for every query. The output therefore has no dependence on the query. This is an architectural invariant, not an optimization failure.

A four-by-four, 77-channel synthetic run confirms all three predictions after 20 updates:

Measurement	Result
Maximum output difference across 16 spatial queries	<code>0.0</code>
Query-embedding gradient norm	<code>0.0</code>
Output/target spatial-variance ratio	<code>0.0</code>

One latent remains a useful negative control, but it is not a viable configuration for a window containing multiple output locations.

Root cause 2: the configured attention path is narrower than the target state

The production task emits 77 prognostic channels per spatial query. The model config sets `queries_dim=64`. At the time of the recorded experiments, `PerceiverConfig.build_io` did not expose the dependency's `cross_heads` or `cross_dim_head`; their defaults were one head of width 64. Increasing `latent_dim` therefore did not widen either cross-attention value path. This branch now exposes and tests both controls for the matched follow-up configuration.

To remove spatial routing, the next experiment uses a two-by-two field decoded as four independent one-token/one-query windows. It trains on a fresh batch of 64 random channel vectors on every update and evaluates 128 unseen vectors after 500 updates. Consequently, static coordinate memorization cannot solve the task.

Latent / query width	Cross-attention inner width	Held-out MSE	Output/target spatial variance
64 / 64	64	0.437818	0.545
128 / 128	64	0.239298	0.745
128 / 128	128	0.021848	0.986
128 / 128	2 heads x 64	0.021848	0.986

The one-head width-128 and two-head width-64 runs are numerically identical for this one-token case. Widening the true attention value path reduces held-out error by 90.9% relative to widening only the latent and query representations, and by 95.0% relative to the current 64-wide path. The widened curve is still decreasing at update 500, so 0.021848 is not interpreted as an asymptotic floor.

This establishes a channel-information bottleneck independent of spatial routing. It also explains why prior “width 128” controls could improve without resolving the failure: the underlying cross-attention head width remained 64.

Root cause 3: unanchored routing admits memorization and learns copying slowly

Multi-latent Perceiver IO is not universally incapable of fitting a copy task. On four fixed four-by-four samples, both 16 and 64 width-64 latents reach roughly $1e-3$ training MSE after 500 updates, while a width-128 model transiently reaches below $1e-6$. The fit is memorization rather than a learned identity operator: the 64-latent model has held-out MSE 1.660140, worse than predicting zero for standard-normal targets.

Attention instrumentation makes the mechanism visible. For each cross-attention stage, the harness records normalized entropy, stable rank, and maximum mass. It also multiplies the head-averaged query-to-latent and latent-to-data attention matrices to obtain an approximate query-to-input transport matrix. Self-attention and value transformations mean this product is not a complete model Jacobian, but its same-cell mass is an interpretable routing diagnostic.

Training task and width	Held-out MSE	Approx. diagonal mass	Approx. top-1 alignment	Output/target spatial variance
Four fixed fields, width 64	1.660140	0.064	0.070	0.998 on training fields
Fresh fields, width 64	0.682370	0.905	0.938	0.287
Fresh fields, width 128	0.300376	0.921	0.938	0.651

The fixed-sample model reconstructs its training fields with diffuse transport. It can encode a sample signature into global latents and use coordinate queries to emit a memorized field, without learning input-cell-to-output-cell correspondence. Fresh random fields prevent this shortcut. They force both widths to discover a mostly diagonal route, but the route emerges only late in optimization and the model still under-reconstructs spatial variance after 300 updates.

This is consistent with the prior ocean controls: increasing unanchored latent count from 1 through 256 did not solve the 12-by-12-window task, whereas reducing the task to one input and one output cell closed most of the gap. The evidence now supports an optimization and inductive-bias diagnosis: a valid multi-latent route exists, but the unordered bottleneck makes learning spatial correspondence unnecessarily difficult and permits misleading fixed-set memorization.

Architectural evidence: direct and position-anchored cross-attention

Two structure-changing controls retain coordinate queries and therefore admit arbitrary output grids:

1. **Direct cross-attention** removes the internal learned latent bank. Output queries attend directly to processor tokens, followed by a query residual, feed-forward block, and 77-channel projection.
2. **Position-anchored direct cross-attention** adds a strong relative-position logit bias to the same learned attention. On a matching grid it starts at the corresponding input cell; on a different output grid the same construction can be evaluated from physical input/output coordinates. Content-dependent logits remain free to learn corrections to the geometric route.

All following models use a 128-wide, one-head attention path and train on fresh random fields. The four-by-four Perceiver IO and plain direct controls use eight training samples per update; the 12-by-12 controls use four because of the larger attention matrix. Held-out evaluation uses respectively 16 and 8 unseen samples.

Architecture	Grid / context	Parameters	Held-out MSE at update 300	Diagonal or max attention mass	Spatial variance ratio
Perceiver IO	4x4 / none	1,040,589	0.300376	0.921 approximate diagonal	0.651
Direct cross-attention	4x4 / none	307,661	0.010044	0.995 diagonal	0.973

Architecture	Grid / context	Parameters	Held-out MSE at update 300	Diagonal or max attention mass	Spatial variance ratio
Direct cross-attention	12x12 / none	307,661	0.500880	0.549 diagonal	0.476
Direct cross-attention	12x12 / one ring	307,661	0.857778	0.399 maximum	0.072
Position-anchored direct	12x12 / none	307,661	0.005325	>0.999999 diagonal	0.986
Position-anchored direct	12x12 / one ring	307,661	0.005244	>0.999999 maximum	0.986

Removing the latent bank improves four-by-four held-out MSE by 96.7%, uses 70.4% fewer parameters, and runs more than twenty times faster in the local CPU probe. It does not by itself solve scale: at 12-by-12 the route is still being discovered at update 300, and irrelevant context makes that discovery harder.

The explicit spatial prior removes the scale and context sensitivity. At 12-by-12 it reduces held-out MSE by 98.9% without context and 99.4% with context, relative to plain direct attention. The context/no-context learning curves are nearly identical because context tokens no longer compete equally with the corresponding source cell. This supports a flexible-resolution decoder built around geometric routing plus learned residual correction, rather than an unordered second latent bottleneck.

Cross-resolution control: anchoring alone is not enough

The hard same-grid anchor above must not be mistaken for proof of useful cross-resolution behavior. A second synthetic task evaluates five smooth analytic latitude/longitude basis fields with independent coefficients for every sample and channel. It observes them on an 8-by-8 source grid and reconstructs the same continuous fields on a 16-by-16 output grid. Training again uses fresh random coefficients and evaluation uses unseen coefficients.

As geometry-only references, direct bilinear interpolation has MSE 0.015970 and nearest-neighbor interpolation has MSE 0.029651; the target variance is 0.488414. Circularly padding longitude before bilinear interpolation lowers the reference to 0.015492, a further 3.0% improvement. The learned pointwise projection reaches 0.015562 at update 300 when trained with a suitable learning rate, consistent with the nonperiodic bilinear floor.

Decoder geometry	Same-grid random MSE	8-to-16 analytic MSE	Interpretation
Sharp position attention, strength 16	0.005244	0.044621	Excellent identity, approximately nearest-neighbor off-grid
Broad position attention, strength 2	0.040001	0.012272	Learns useful interpolation, weakens exact identity
Bilinear resampling + projection	0.0000148	0.015562	Stable identity and predictable interpolation
Bilinear base + zero-initialized attention residual	0.000000196	0.012608	Preserves identity and improves the interpolation base

The attention-only results demonstrate a real temperature tradeoff: a kernel sharp enough to enforce identity cannot interpolate, while a kernel broad enough to interpolate allows neighboring tokens to contaminate same-grid copying. The hybrid removes that conflict. Its synthetic base projection is initialized to copy the 77 aligned input channels, so the absolute same-grid number is an oracle diagnostic, not a claim that processor features already equal output channels. The relevant architectural result is that a zero-initialized residual cannot damage the base at initialization and subsequently reduces cross-resolution error by about 21%.

The production analogue should therefore be:

1. resample the canonical processor grid deterministically to the requested output coordinates, using periodic longitude and coordinate-aware latitude rather than edge-clamped index interpolation;
2. project channels with the learned pointwise head already validated by the ocean proxy;
3. add a zero-initialized local attention residual whose logits include relative physical position, periodic longitude, and an explicit validity mask.

This preserves the successful resampling decoder exactly at initialization, keeps output-grid flexibility, and gives the model a route to learn corrections where bilinear interpolation is inadequate. This was the synthetic-stage recommendation; the later wet-mask intervention supersedes steps 1–2 for production by projecting physical channels on the native grid before channel-specific resampling, and the learned-encoder screen rejects step 3 unless a new residual defect appears.

Learned-encoder S0 confirmation

The earlier oracle-copy probe placed target channels directly in the decoder input. The follow-up S0 screen removes that shortcut: a two-layer pointwise encoder learns the 16-to-32-channel representation jointly with each decoder from fresh analytic coefficients, and evaluation uses 256 unseen coefficient draws. The table reports three-seed held-out means after 2,000 Adam updates at learning rate $3\text{e-}3$.

Decoder	Same 8-to-8	Down 16-to-8	Shifted 8-to-8	Up 8-to-16
Physical-coordinate resampling + projection	0.000182	0.000208	0.004716	0.003459
Physical base + zero-init local attention	0.000301	0.000326	0.004890	0.003487
Position-anchored attention only	0.001035	0.000652	0.005568	0.006257

The preceding 500-update screen also isolated coordinate semantics. Shape-only bilinear interpolation tied physical interpolation on the identical grid, but was 10.0 times worse when downsampling, 8.7 times worse on the half-cell longitude shift, and 3.6 times worse when upscaling. This is direct evidence that the dominant flexible-resolution architectural fix is physical-coordinate resampling, not more attention capacity.

The longer confirmation revises the earlier hybrid recommendation. Its correction helped same-grid and down-grid error at 500 updates, but the simpler learned base continued improving and won every route by 2,000 updates. The hybrid was also about 4–7 times slower on CPU. Under the plan’s successive-halving rule, it does not advance as a co-equal S1

candidate; its checked-in implementation remains a fallback if ocean spectra expose a residual that a pointwise projection cannot learn.

Position-only anchored attention was not rejected because it cannot express the desired mapping. It was rejected as the primary renderer because, under the same learned-encoder budget, it lost every route, required about 8–10 times the runtime of the base, suppressed shifted-grid high-wavenumber power to 0.815, and amplified upscaled high-wavenumber power to 1.264. It remains an informative control, while the zero-initialized hybrid is the safer way to add it if later evidence justifies the cost.

Real-ocean learned-inverse confirmation

S1 tests the promoted physical-coordinate resampling decoder with the production two-layer Perceiver encoder on 154-channel, masked one-degree OM4 states. The processor is bypassed, but neither the encoder nor the decoder projection is given an identity initialization. Every result is measured on 32 held-out states that are disjoint from the 32 fixed training states.

Encoder geometry	LR	Best held-out MSE	Final MSE	Mean high-k ratio	Mean amplitude ratio	Mean absolute bias / target std
additive	3e-4	0.021202	0.021202	0.8851	0.9561	0.0092
additive	6e-4	0.015452	0.020102	0.9264	0.9784	0.0125
additive	1e-3	0.013569	0.015551	0.9325	0.9814	0.0118
none	3e-4	0.020776	0.020776	0.8372	0.9310	0.0149
none	6e-4	0.013780	0.013780	0.9247	0.9745	0.0098
none	1e-3	0.011844	0.011844	0.9283	0.9758	0.0076

At the useful 1e-3 learning rate, omitting post-encoder additive position/scale embeddings wins every variable-group MSE and lowers mean absolute normalized bias. The selected model’s high-wavenumber ratios are 0.929 for temperature, 0.953 for salinity, 0.894 for zonal velocity, 0.929 for meridional velocity, and 1.094 for SSH; all satisfy the preregistered [0.85, 1.15] gate. The low-rate result prevents a blanket claim that geometry removal always helps: this is an optimization-by-geometry interaction, with a clear advantage at the learning rate that best trains both variants.

This result supports a learned inverse, not a dumb encoder. The encoder still must compress each masked ocean state into the canonical latent representation. Only the decoder’s spatial correspondence is anchored by physical-coordinate resampling; its channel projection remains learned. Geometry is also retained as a sidecar for the processor experiments, where a nonzero-depth latent evolution may need it even though adding it directly to depth-zero encoder features hurts reconstruction.

The selected 0.011844 is consistent with closing the earlier decoder gap, but it is not treated as a strict head-to-head improvement over the historical direct/direct 0.012083: the harnesses and held-out sample contracts differ. The controlled conclusion comes from the six S1 runs above. Width {256, 380} and latent-dimension 128 confirmations then isolated whether more internal encoder capacity or a wider processor interface was preferable.

Embedding width	Latent dimension	Best held-out MSE	Final MSE	Mean high-k ratio	Mean amplitude ratio	Mean absolute bias / target std
128	64	0.011844	0.011844	0.9283	0.9758	0.0076
256	64	0.010375	0.010903	0.9378	0.9798	0.0223
380	64	0.009505	0.009505	0.9703	0.9956	0.0221
128	128	0.007937	0.007937	0.9256	0.9707	0.0135

The 128/128 model wins aggregate MSE and every variable group. Its high-k ratios are 0.950 for temperature, 0.941 for salinity, 0.889 for zonal velocity, 0.919 for meridional velocity, and 0.987 for SSH. This selects extra internal encoder capacity rather than a wide external state: the canonical processor interface remains 128 channels, while the Perceiver encoder latent dimension rises to 128. The zero-depth parameter count changes only from about 0.92M to 0.97M over the width sweep, but a 380-channel interface would make every repeated processor application substantially wider.

Multi-resolution root cause: encoder spatial bandwidth

The first balanced one/half-degree ocean control uses one shared Perceiver encoder and physical-resampling decoder for all four input/output routes. It completed 40 epochs (1,280 batch-one updates) with one learned vector per one-degree patch. The resolved runtime was naive attention with bfloat16; the run suffix incorrectly says fp32. Stability through completion shows that bfloat16 alone does not reproduce the earlier flash/bfloat16 L40S fault.

Route	Held-out MSE	Mean high-k ratio	Patch-seam ratio
1 degree -> 1 degree	0.01954	1.158	undefined
1 degree -> 1/2 degree	0.06027	0.508	0.898
1/2 degree -> 1 degree	0.01832	1.066	undefined
1/2 degree -> 1/2 degree	0.05737	0.496	0.875

The failure follows output bandwidth rather than cross-resolution direction. Both half-degree outputs retain only about half the target high-wavenumber power, including the half-to-half route where decoder source and output coordinates are already identical. The one-to-half MSE reaches the deterministic interpolation scale while retaining the same coarse spectrum. Thus the physical decoder is doing the expected rendering; the encoder has already compressed four native half-degree cells into one one-degree token.

This is evidence against spending the next run on decoder attention. It also reveals a limitation in the checked-in packed spatial-query encoder: ordered query outputs are stored as channels at a coarse patch center, but the shared pointwise decoder has no intra-patch coordinate with which to select them. With one internal latent, those PerceiverIO output queries are query-blind as well. A viable query version must spatially unpack outputs onto a finer canonical grid and use multiple or directly anchored encoder tokens.

The simpler control now projects every native input cell into learned latent channels and physically resamples those features onto the finest configured grid. It uses ordinary learned initialization, carries no prognostic bypass, and fixes the latent grid independently of the

requested output. It therefore still forms a learned representation that can be processed zero to N times.

One-degree encoder	External width	Held-out MSE	Mean high-k ratio	Mean amplitude ratio	Mean absolute bias / target std
Perceiver, latent dim 128	128	0.007937	0.9256	0.9707	0.0135
Canonical pointwise/resample	128	0.002857	0.9677	0.9965	0.0020
Canonical pointwise/resample	160	0.002197	0.9728	0.9971	0.0018

Both canonical-grid controls decisively beat the Perceiver encoder at the same zero-depth task. Width 160 wins every variable group and is the active matched four-route candidate; width 128 remains the lower-cost processor fallback.

The fixed-finest-grid four-route run then provided a useful falsification by epoch 10. Half-to-half MSE fell from the coarse control's final 0.05737 to 0.00821, and its high-k ratio rose from 0.496 to 0.968. But one-to-one MSE was 0.03178 with high-k ratio 0.446. Upsampling native one-degree features to a half-degree canonical grid and linearly sampling them back is a smoothing operation, not an inverse. Because this is a spatial invariant of the fixed-grid transport, the run was cancelled after 320 updates rather than completing the remaining 960.

The revised candidate retains the learned pointwise channel representation on the native input grid. The shared processor remains fully convolutional and can run the same weights zero to N times at any supported input resolution; only the decoder resamples to the requested output coordinates. This is target-independent, keeps both same-grid spatial paths exact, and avoids both measured bottlenecks. It does trade constant processor resolution for native-resolution compute, which quarter-degree memory and throughput tests must quantify.

The first deterministic half-to-one baseline was itself invalid: physical zero was used where a destination wet cell had no wet source neighbor, yielding normalized MSE near 39.4. The corrected diagnostic uses destination climatology at unsupported locations while retaining mask-renormalized physical interpolation elsewhere. This does not change any learned model result above. The corrected normalized references are 0.005429 for half-to-one, 0.061161 for one-to-half, and numerical zero for same-grid routes.

The native-grid candidate then completed all 1,280 matched updates:

Route	Held-out MSE	Mean high-k ratio	Physical resampler MSE
1 degree -> 1 degree	0.00358	0.992	0
1 degree -> 1/2 degree	0.06120	0.438	0.06116
1/2 degree -> 1 degree	0.02934	0.621	0.00543
1/2 degree -> 1/2 degree	0.00348	1.000	0

This closes the encoder-bandwidth diagnosis: both same-grid routes reconstruct accurately with essentially exact spectral amplitude, and one-to-half reaches its irreducible interpolation error. The remaining asymmetric half-to-one error is a different contract failure. One contributor is that each product is standardized with different scalar channel means and

standard deviations, while the native encoder and resampling decoder are shared affine 1-by-1 maps. Those maps commute with unmasked interpolation but cannot be identity maps in two different normalized coordinate systems and also apply a route-dependent source-to-target affine transform.

The first matched control therefore uses one common channel-normalization basis across native products and also reports a source-normalized interpolation baseline. At epoch 10 it reduces half-to-one MSE from 0.03459 to 0.03013, but remains far above the 0.00543 reference. This does not support normalization as the sole root cause.

A checkpoint-only decomposition makes the scale of that effect explicit. With the original independent statistics, half-to-one interpolation performed directly in the source-normalized basis has MSE 0.00971; the physical-to-target-normalized reference is 0.00543. Thus normalization accounts for about 0.00428, while the trained model is 0.02391 above the physical reference. One-to-half similarly changes from 0.06116 in the physical reference to 0.05620 in the source-normalized control. The much larger remaining learned error requires a second mechanism.

The stronger structural mismatch is mask ordering. The baseline encoder mixes prognostic and boundary channels before the decoder interpolates 160 latent channels. Ocean validity is instead channel dependent: depth levels and variables have different wet masks, and the physical reference independently renormalizes the four interpolation weights for each of 154 output channels. Interpolation is linear for a fixed mask, but these different per-channel denominators do not commute with learned channel mixing. Same-grid routes conceal the problem because no interpolation occurs.

The next matched control decodes latent features to prognostic channels on the native source grid, then performs physical-coordinate interpolation with the immutable channel-wise input masks. It retains a learned representation, does not copy prognostic values around the encoder, supports arbitrary output coordinates, and leaves the shared processor callable zero to N times. Its implementation is the `project_before_resample` path in `ResampleProjectionDecoder`; the data path carries source geometry through `GridContext.input_mask`.

That ordering has now been isolated with an exact checkpoint swap:

Route	Latent-resample baseline	Project then channel-mask resample	Change
1 degree -> 1 degree	0.00513708	0.00513707	< 4e-9
1 degree -> 1/2 degree	0.0877352	0.0867701	-0.000965
1/2 degree -> 1 degree	0.0260986	0.00996908	-0.016130
1/2 degree -> 1/2 degree	0.00564119	0.00564118	< 2e-9

Both evaluations use the exact epoch-25 common-statistics weights; no optimizer step occurs. Relative to the half-to-one physical floor 0.00542936, the ordering change removes 78.0% of excess error while leaving both same-grid functions unchanged. This is direct evidence that per-channel wet-mask interpolation cannot be reconstructed after latent channel mixing. The remaining 0.00454 excess is similar in scale to the learned same-grid projection error and is being tested with a fresh matched optimization.

The fresh run has passed its epoch-5 bring-up gate. Half-to-one is already 0.02121, compared with 0.03737 for the common-statistics latent-resampling control at epoch 5, while same-grid errors remain comparable. Aggregate MSE improves from 0.04350 to 0.03856. At epoch 5, the channel-wise masked training path costs 156 seconds for 32 samples versus 71 seconds for the control, an observed 2.2x training-throughput penalty. `identity/epoch_seconds` excludes the subsequent route evaluation, whose cost must be measured separately before quarter-degree training. Explicit training, evaluation, and total-duration metrics are now emitted for subsequent runs.

By epoch 10, half-to-one reaches 0.01352 versus the control's matched 0.03013; same-grid errors are 0.00877 and 0.00993. Aggregate MSE 0.03063 is already below the latent-resampling control's epoch-40 result 0.03079, establishing that the corrected ordering improves optimization as well as the checkpoint swap.

The epoch-15 result strengthens that conclusion: aggregate MSE is 0.02830, half-to-one is 0.01121 versus its 0.00543 deterministic floor, and same-grid MSEs are 0.00634 and 0.00726. One-to-half is 0.08839 versus its 0.07900 floor. Thus the remaining error is increasingly comparable to the learned same-grid projection error, rather than the much larger latent mask-order failure.

At epoch 20, fresh half-to-one MSE is 0.01003, effectively reproducing the checkpoint-only order swap's 0.00997. Same-grid errors are 0.00509 and 0.00588, one-to-half is 0.08736, and aggregate MSE is 0.02709. Agreement between the intervention on fixed weights and independent optimization is strong evidence that projection-before-channel-masked-resampling addresses the causal failure rather than merely favoring one training trajectory.

The fresh run finishes at its epoch-40 validation best. Aggregate MSE is 0.02491; route MSEs are 0.00304 for one-to-one, 0.08498 for one-to-half, 0.00805 for half-to-one, and 0.00357 for half-to-half. Half-to-one is only 0.00262 above its 0.00543 deterministic floor. Same-grid and half-to-one high-k ratios are 0.936, 0.938, and 0.934, so the accuracy gain does not come from spectral smoothing.

Quarter-degree zero-shot evidence

The epoch-10 one/half-degree checkpoint was evaluated without optimization on all nine routes after adding the unseen quarter-degree grid:

Route	Model MSE	Deterministic MSE	Excess	High-k ratio
1 -> 1	0.00865	0.00000	0.00865	0.899
1 -> 1/2	0.08824	0.08265	0.00559	0.551
1 -> 1/4	0.10897	0.10903	-0.00006	0.651
1/2 -> 1	0.01348	0.00592	0.00756	0.888
1/2 -> 1/2	0.00981	0.00000	0.00981	0.886
1/2 -> 1/4	0.03747	0.02959	0.00788	0.635
1/4 -> 1	0.07040	0.07933	-0.00892	1.351
1/4 -> 1/2	0.01158	0.00268	0.00890	0.874
1/4 -> 1/4	0.01055	0.00000	0.01055	0.848

Quarter same-grid error is comparable to the trained resolutions, and the two coarse-to-quarter routes remain close to their deterministic floors. The learned inverse therefore generalizes to a new native grid without a finest-grid query bypass or scale embedding in the encoder.

Quarter-to-one reveals a new transport-specific failure rather than an encoder failure. Its aggregate high-k ratio 1.35 includes velocity at 1.73 ; bilinear point sampling aliases a fourfold restriction, while the target OM4 products were conservatively regridded. The promoted architecture should keep native-grid learned projection followed by channel-masked transport, but use conservative or antialiased restriction for substantially coarser outputs and bilinear coordinate prolongation for finer outputs.

The fixed-checkpoint area diagnostic isolates the restriction effect:

Route	Bilinear floor MSE	Area floor MSE	Model high-k	Area high-k
1/2 -> 1	0.00592	0.00741	0.888	0.997
1/4 -> 1/2	0.00268	0.00399	0.874	1.010
1/4 -> 1	0.07933	0.01123	1.351	1.001

Area averaging is decisive for 4x restriction: MSE is 86% lower than bilinear and high-k power is essentially exact. At 2x it restores high-k power but raises MSE, so the production operator should be scale aware. Use a physically weighted conservative or antialiased restriction for large ratios; retain bilinear prolongation; and test a better low-pass kernel before replacing bilinear at 2x.

The unchunked evaluator took 58 minutes after container setup and held about 52 GiB of GPU memory in a spot check. The exact chunked eval-only rerun matches all nine route MSEs and aggregate 0.0399052 , takes 1,162.5 seconds for its sole held-out pass, and peaks at 10.30 GiB allocated / 24.43 GiB reserved CUDA memory. Its cold Slurm wall time is 56:20 because SIF construction takes about 36 minutes; the warm area control completes in 15:54. The bounded-memory flexible- resolution path therefore passes its engineering gate. The clean and area jobs are Slurm 14574139 / W&B pod127b1 and Slurm 14579543 / W&B rqlh7bke .

Historical processor-depth evidence (refinement control only)

The numbers in this section predate the physical-time clarification. They compare repeated applications against one common current-state/refinement target. Do not use them to select a physical latent-autoregressive rollout.

Matched 1,280-sample runs compare forecast-only processor training against the source-grid inverse regularizer while decoding the same latent at every depth:

zero-depth weight	depth 0	depth 1	depth 2	depth 4
0	0.024406	0.022151	0.033715	0.135192
0.05	0.021861	0.022284	0.030805	0.113853

The untouched exact-window checkpoint is 0.024567 . Weight 0.05 improves the inverse by 11.0%, is effectively tied at depth one, and improves depths two/four by 8.6%/15.8% over weight zero. It is the selected training mixture. But depth four remains 5.2x depth zero:

training only one processor call, even with eight times the smoke's samples, does not teach stable arbitrary iteration. The next causal test is shared-weight training with positive depths sampled from {1,2,4}; do not constrain the encoder or add decoder attention to address this processor- specific failure. The runs are Slurm 14581788 / W&B s8rig9hd and Slurm 14581789 / W&B ulkilobr .

The matched multi-depth test is now complete (Slurm 14590678 , W&B 9yart3ax):

training depths	depth 0	depth 1	depth 2	depth 4
fixed {1}	0.021861	0.022284	0.030805	0.113853
cycled {1,2,4}	0.022202	0.021475	0.021805	0.024028

Cycling depths is the only causal change. It removes 29.2% of depth-two error and 78.9% of depth-four error while changing depth zero by +1.6% and improving depth one by 3.6%. Depth four is only 8.2% worse than depth zero, and all four resolution routes improve at depth four. The failure was inadequate training support for the requested iteration counts, not evidence that the learned encoder must be made trivial, that the decoder needs attention, or that the processor requires a global residual. Promote this checkpoint to forecasting; retain residualizing the processor only as a falsifiable fallback if physical-time training drifts again.

Historical forecast proxy and post-forecast refinement evidence

The one-step forecast comparisons remain evidence that the selected representation can support a processor. The repeated-depth comparisons are mis-specified for the desired $t+N$ contract because depths two and four were also paired with $t+1$.

The first calibrated one-degree forecast proxy starts from that multi-depth checkpoint and then trains only processor depth one. Two seeds at each inverse weight use the same 512 stratified samples per epoch, 12 epochs, 192 optimizer updates, and effective global batch 32:

zero-depth inverse weight	seed 15 validation MSE	seed 16 validation MSE	mean
0	0.0315724	0.0315876	0.0315800
0.05	0.0316064	0.0315465	0.0315765
0.2	0.0315684	0.0316211	0.0315947

The regularizer is forecast-neutral: all three means fall within 0.06%. Every arm beats the matched v2 proxy 0.042390 by about 25.5%, the historical Perceiver/resampling proxy 0.051655 by 38.8–38.9%, and the direct one-cell proxy 0.041278 by about 23.5%. Weight 0.2 also gives the best velocity spectral retention: its two-seed zonal high-k ratio is about 0.733 , versus about 0.718 at 0.05 and 0.703 without the inverse term. This independently supports the selected encoder/decoder path as a forecast representation rather than an autoencoder-only shortcut.

Fixed-depth forecast training nevertheless erases the 0...N refinement behavior. On the same cross-resolution current-state window, post-forecast depth-zero MSE is 0.0374764 without the inverse loss and 0.0281127 with weight 0.05, averaged over the two seeds. Relative to the pre-forecast checkpoint's 0.0222023 , these are regressions of 68.8% and 26.6%. The inverse term is effective but the 0.05 arm narrowly misses the provisional 25% preservation gate.

A separate held-out $t+1$ audit distinguishes the inverse from forecast semantics:

zero-depth inverse weight	depth 0	depth 1	depth 2	depth 4
0	0.0137334	0.0478494	0.0968079	0.291519
0.05	0.00631293	0.0476432	0.0880912	0.219623

Depth zero is a persistence-like control against the future target and depth one is the trained one-step prediction. Repeating the processor toward the same refinement target degrades at depths two and four for every seed. This is the same causal exposure failure found in identity training, now reproduced after forecast fine-tuning. It is not evidence against the learned encoder or mask-aware decoder. Regular training therefore now supports a default-off `train_processor_depths` cycle, and the selected proxy/full configs train and validate physical leads `[1,2,4]`.

The stronger fixed-depth weight-0.2 inverse audit passes the preservation gate: its two-seed depth-zero mean is `0.0241583`, only 8.8% worse than the pre-forecast checkpoint. More importantly, the matched multi-depth forecast rerun is complete:

run	seed 15	seed 16	mean
fixed depth 1, weight 0.2	0.0315684	0.0316211	0.0315947
cycled depths <code>{1,2,4}</code> , weight 0.2	0.0360968	0.0361911	0.0361440

The depth cycle costs 14.4% proxy MSE at the fixed 192-update budget, but the result still beats matched v2 by 14.7%, direct one-cell by 12.4%, and the old Perceiver/resampling proxy by 30.0%. It also restores the iteration contract on the true $t+1$ audit:

multi-depth forecast	depth 0	depth 1	depth 2	depth 4
two-seed mean	0.00371387	0.0413635	0.0455987	0.0485074

Depth four is now only 17.3% worse than depth one rather than roughly 4.6 times worse after fixed-depth training. The post-forecast current-state inverse mean is `0.0246888`, an 11.2% regression from the pre-forecast checkpoint and comfortably inside the 25% gate. This historically promoted inverse weight `0.2` with positive depths `[1,2,4]` only for the same-target refinement formulation; it is not promoted for physical latent autoregression. The corresponding W&B runs are `9i47kib4` and `t7855wfw`; forecast-depth audits are `dp1y2chf` and `3yluahc9`; inverse audits are `142he1n3` and `hufdi48y`.

Full-scale physical latent-autoregression evidence

The full one-degree result is summarized in the executive conclusion. The balanced one/half-degree proxy then tests all four physical routes with the same frozen inverse and ReZero transition for 192 optimizer updates:

Route	Lead 1 MSE / persistence reduction	Lead 2	Lead 4
1 -> 1	<code>0.045609 / 45.5%</code>	<code>0.076865 / 49.1%</code>	<code>0.103631 / 53.8%</code>
1 -> 1/2	<code>0.108017 / 22.3%</code>	<code>0.134762 / 29.3%</code>	<code>0.163326 / 34.5%</code>
1/2 -> 1	<code>0.041871 / 52.7%</code>	<code>0.069580 / 55.4%</code>	<code>0.095643 / 58.3%</code>

Route	Lead 1 MSE / persistence reduction	Lead 2	Lead 4
1/2 -> 1/2	0.056481 / 52.3%	0.096918 / 54.3%	0.130894 / 57.4%

At lead four, zeroing the boundary raises proxy route MSE by 19.5--33.1%, while reversing boundary order raises it by 14.3--39.7%. The weaker and occasionally negative lead-two time-reversal response was a reason to retain these controls in the full run rather than declaring boundary causality solved.

The full-interval run is Slurm job 14666431 and W&B 2ruaaimh. It completes all 17 epochs and 6,392 optimizer updates on six RTX6000s. Epoch 17 is both validation-selected and terminal:

Route	Lead 1 MSE / persistence reduction	Lead 2	Lead 4
1 -> 1	0.026417 / 68.5%	0.042275 / 72.0%	0.054706 / 75.6%
1 -> 1/2	0.083557 / 39.9%	0.096616 / 49.3%	0.108374 / 56.5%
1/2 -> 1	0.022313 / 74.8%	0.032732 / 79.0%	0.041997 / 81.7%
1/2 -> 1/2	0.027011 / 77.2%	0.044697 / 78.9%	0.058734 / 80.9%

Every route beats its lead-matched persistence baseline at every lead. Unlike the proxy, every route also responds in the expected direction to both forcing controls at every lead. Aggregate zero-boundary increases are {23.4%, 61.1%, 124.7%}, and time-reversal increases are {9.8%, 19.2%, 93.7%}. The frozen same-grid inverse remains bitwise constant through training.

Slurm job 14687084 loads that selected checkpoint in validation-only mode and performs no optimizer or checkpoint write. Its independent scalar results agree within 6e-7, and W&B ke7l2j7a records the endpoint route spectra:

Route	Temperature	Salinity	Zonal velocity	Meridional velocity	SSH	Defined seam range
1 -> 1	0.969	0.975	1.048	1.110	1.021	undefined
1 -> 1/2	0.938	0.959	0.288	0.149	0.977	0.811--1.002
1/2 -> 1	0.963	0.966	0.884	0.872	1.006	undefined
1/2 -> 1/2	0.963	0.978	0.790	0.751	1.025	1.000--1.002

One-degree outputs have no within-patch edge against which to normalize a seam, so that metric is undefined rather than failed. Coarse-to-fine velocity power is an input-information limit. The half-degree same-grid velocity deficit and the 28.7% one-to-one lead-one regression relative to the single-resolution model are processor/exposure limitations to resolve before quarter-degree training; the frozen decoder cannot cause their training-time drift.

Architecture decision matrix

Candidate	Same-grid identity	Flexible output grid	Learned nonlocal correction	Evidence-backed decision
Current Perceiver IO	Poor at production scale	Yes	Yes	Do not promote unchanged

Candidate	Same-grid identity	Flexible output grid	Learned nonlocal correction	Evidence-backed decision
Wider Perceiver IO	Channel bottleneck reduced	Yes	Yes	Do not promote: it fixes only the measured channel bottleneck, not routing
Direct query-to-token attention	Better at 4x4, slow at 12x12	Yes	Yes	Useful ablation, not sufficient alone
Position-only anchored attention	Learnable but slower and less accurate	Yes	Yes	Retain as a control; do not promote as sole renderer
Physical-coordinate resampling projection	Best learned-encoder S0 result	Yes	No	Selected transport base, with projection moved before masked resampling
Resampling + zero-init attention residual	Preserves base at initialization	Yes	Yes	Retain as fallback; S0 does not justify its added cost
Native-grid latent resampling + learned projection	Excellent same-grid inverse and spectra	Yes	No	Baseline: cross-grid error exposes normalization and mask-order contracts
Native-grid learned projection + channel-masked resampling	Exact same-grid transport by construction	Yes	No	Selected learned inverse decoder
Frozen inverse + latent ReZero transition	Preserves the learned inverse exactly during forecast training	Yes	Yes, in the processor	Selected contract; tune processor exposure before quarter degree

The selected decoder does not send reconstructive values through LayerNorm.

`ResampleProjectionDecoder` applies its learned pointwise channel map to raw native-grid latent values, then transports each physical output channel with its own wet mask. LayerNorm is confined to attention controls that were not selected. The learned state encoder is likewise not initialized or constrained to be an identity: the reconstruction contract is the composition $D(E_{\text{state}}(x))$, not either component alone. Freezing that learned pair during forecast training makes processor depth zero the measured inverse, while zero or more shared processor calls remain legal and decoding can target any supplied coordinate grid.

The glossary above provides the config surface, equations, and direct source links for every selected modification and retained control. In particular, the runnable `anchored-cross-attention` probe uses `CoordinateAttentionDecoder` backed by `LocalCoordinateAttentionCorrection`; `AnchoredCrossAttentionIO` is a retained lower-level square-window prototype, not the class selected by that CLI route. Rollout ownership and latent chunk carry are implemented by `BaseModel.initialize_rollout`, `SamudraMulti.inference`, and `run_rollout`.

The same `SamudraMulti.reconstruct_once` path now constructs a source-grid context for the zero-depth MSE. Consequently the inverse regularizer is independent of the chosen forecast output resolution, while the processor geometry sidecar remains outside the encoder and is applied once per processor iteration.

Channel-conditioned queries are not the first remedy. They would multiply query count by up to 77 while the current spatial routing failure remains. Producing all channels from one spatial query is reasonable once the query/value path is at least as wide as the state; variable-group heads may be worth testing later if velocity continues to lag after routing is fixed.

Limitations and remaining evidence

- The first L40S ocean bring-up isolated a CUDA illegal-memory access in the bfloat16/flash backward both with and without activation checkpointing. The completed fresh ocean run validates naive attention with bfloat16, but the flash backend remains a production-runtime limitation to diagnose separately.
- Separate RTX6000 profiles of the selected U-Net latent transition fail in the first backward at per-rank batches 4, 8, and 16 with a CUDA illegal-address error, even where the reported allocation estimate is only 4.03 GiB. The full one/half-degree validation instead uses six ranks, per-rank batch one, and accumulation five for effective batch 30 and exactly 6,392 scheduled updates. The failed profiles make no optimizer step and are runtime evidence, not model evidence.
- The multiplied attention matrices are a routing diagnostic, not an exact model Jacobian because self-attention and value transformations intervene.
- The production checkpoint swap and fresh run confirm the mask-order mechanism on actual OM4 coordinates, periodic longitude, nonuniform latitude, and channel-specific wet masks. Quarter-degree zero-shot scaling and the bounded-memory rerun are complete. The fixed area reference establishes that restriction must be scale aware: it is decisive at 4x but not a universal 2x replacement. A spherical conservative restriction prototype is checked in and unit-tested, but its checkpoint audit was cancelled before optimization or evaluation when the review boundary moved to the one/half-degree result. It is therefore deferred and must not be treated as selected architecture.
- The hybrid synthetic control initializes its base projection to copy aligned target channels. Real processor features require a learned channel projection. The completed ocean resampling proxy is the evidence that such a projection is trainable in the real model.
- S2 confirms the learned inverse across independently regridded one/half-degree products. The corrected state-only inverse now improves every matched route and zero-shot quarter same-grid MSE by 87% relative to the earlier joint checkpoint. The corrected physical latent-autoregressive proxy selects the frozen ReZero transition, and the full-data one-degree validation completes with the inverse unchanged. The full one/half-degree validation and selected-checkpoint endpoint audit are complete with the inverse unchanged.
- Small forecast controls use two seeds, but each full-data promotion run uses one seed. The one/half-degree endpoint is stable and independently reproduced by the audit, but that is checkpoint reproducibility rather than training-seed uncertainty.
- The full multi-resolution model misses the provisional one-degree same-route degradation gate (28.7% versus a 25% allowance), and half-degree same-grid velocity power is 0.790/0.751. These are explicit reasons to stop for review rather than promote directly to quarter-degree training.

Reproduction

The core local controls are reproducible from the repository root:

```
# Exact one-latent query blindness.
uv run python scripts/probe_perceiver_decoder.py \
  --grid-size 4 --samples 4 --steps 20 --num-latents 1
```

```
# One-cell channel bottleneck: current true width versus 128-wide attention.
uv run python scripts/probe_perceiver_decoder.py \
  --grid-size 2 --window-patches 1 --samples 64 --eval-samples 128 \
  --steps 500 --fresh-batches --num-latents 1 --latent-dim 64 \
  --queries-dim 64 --cross-dim-head 64
uv run python scripts/probe_perceiver_decoder.py \
  --grid-size 2 --window-patches 1 --samples 64 --eval-samples 128 \
  --steps 500 --fresh-batches --num-latents 1 --latent-dim 128 \
  --queries-dim 128 --cross-dim-head 128

# Production-window spatial anchor.
uv run python scripts/probe_perceiver_decoder.py \
  --architecture anchored-cross-attention --grid-size 12 \
  --context-patches 1 --samples 4 --eval-samples 8 --steps 300 \
  --fresh-batches --queries-dim 128 --cross-dim-head 128

# Flexible 8-to-16 interpolation base plus learned residual.
uv run python scripts/probe_perceiver_decoder.py \
  --architecture resample-attention-residual --data-mode analytic \
  --grid-size 8 --output-grid-size 16 --samples 4 --eval-samples 8 \
  --steps 300 --fresh-batches --queries-dim 128 --cross-dim-head 128 \
  --position-bias-strength 2 --learning-rate 0.003
```

The follow-up real-data control is fully specified by `configs/samudra_multi_om4/identity_1deg_decoder_wide.yaml` and runs through `python -m samudra.identity`.

Exact true-lead and route metrics from a W&B run can be reproduced without destination-grid pooling:

```
uv run python scripts/summarize_latent_ar_runs.py \
  ocean_emulators/default/<run-id> --format json \
  --route 180x360_to_180x360 \
  --route 180x360_to_360x720 \
  --route 360x720_to_180x360 \
  --route 360x720_to_360x720
```

The default selects the lowest lead-one row; add `--selection terminal` to reproduce the terminal row separately.

Recommended follow-up validation

1. Retain the completed width-160 native-grid encoder/decoder, no additive encoder geometry, and projection-before-channel-masked-resampling selection.
2. Do not add the zero-initialized attention residual: neither S2 nor quarter zero-shot transfer exposes a residual defect that justifies its cost.
3. Retain the completed state-only inverse and physical-lead `{1,2,4}` result: freeze the inverse and use a zero-initialized latent transition residual. The inverse-weight sweep shows that a soft reconstruction penalty does not prevent head co-adaptation; do not carry weight 0.2 forward as a conclusion.

4. Separate the selected inverse from the processor promotion decision. Before a larger-resolution run, test whether more per-route updates or modest processor capacity recover the one-to-one degradation gate and half-degree velocity spectra while the frozen reconstruction remains exact.
5. Retain the completed bounded-memory quarter evaluator. After review of this report, validate the deferred scale-aware conservative restriction prototype at 4x and compare a better low-pass kernel at 2x; fixed area averaging lowers the 4x floor by 86% but improves spectra while worsening MSE at 2x.
6. Do not begin a larger quarter-degree training run from this report alone. First require the checkpoint-only restriction audit and report every physical route separately, including persistence, spectra, seam/edge diagnostics, and forcing ablations; destination-grid pooled metrics are not sufficient evidence for flexible output resolution.